

SCIENCE FOR TECHNOLOGY

Short Notes 01: Area and Volume Formulas & Principles

PART 1: 2D SHAPES & SURFACE AREAS

01. Area of a Square

$$\text{Area} = a \times a = a^2$$

Where a is the length of a side.

02. Area of a Rectangle

$$\text{Area} = a \times b = ab$$

Where a is length and b is breadth.

03. Area of a Parallelogram

$$\text{Area} = AB \times h_1 = AB \cdot h_1$$

Types: Oblique/Right Parallelogram, Rhombus, Rhomboid, Square, Rectangle.

04. Area of a Trapezium

$$\text{Area} = \frac{1}{2} \times (a + b) \times h$$

Where a, b are parallel sides and h is perpendicular height.

05. Area of a Circle

$$\text{Area} = \pi r^2$$

Where r is the radius of the circle.

06. Area of a Triangle

- Standard: $\text{Area} = \frac{1}{2} \times \text{Base} \times h = \frac{1}{2} \times BC \times h$
- Right-Angled: $\text{Area} = \frac{1}{2} \times AB \times BC$
- Trigonometric: $\text{Area} = \frac{1}{2} \times a \times b \times \sin(C)$

07. Surface Area of a Cube

$$\text{Total Area} = 6(a \times a) = 6a^2$$

Consists of 6 identical square faces.

08. Surface Area of a Cuboid

$$\text{Area} = 2(ab + ac + bc)$$

Where a, b, c are length, width, and height.

09. Surface Area of a Pyramid

- Square Base:

$$\text{Total Area} = 4\left(\frac{1}{2} \times a \times h\right) + a^2$$

- Rectangular Base:

$$\text{Total Area} = 2\left(\frac{1}{2} \times a \times h_1\right) + 2\left(\frac{1}{2} \times b \times h_2\right) + ab$$

10. Surface Area of a Cone

- Base Area: πr^2
- Curved Area: $\pi r l$
- Total Area: $\pi r^2 + \pi r l = \pi r (r + l)$

11. Surface Area of a Cylinder

- Curved Area: $2 \pi r h$
- Two Circular Ends: $2 \pi r^2$
- Total Area: $2 \pi r^2 + 2 \pi r h = 2 \pi r (r + h)$

12. Surface Area of a Sphere

$$\text{Total Surface Area} = 4 \pi r^2$$

Where r is the radius.

PART 2: CONCEPTS & PRINCIPLES OF VOLUME

- **Definition:** The amount of space an object occupies in three-dimensional space is defined as the **volume** of that object.
- **Hollow Objects:** The volume of the material used to construct a hollow object is strictly less than the total external volume of the object.
- **Solid Objects:** For solid objects, the total volume of the object and the volume of the material contained within it are equal.
- **Uniform Cross-sections (Prisms/Cylinders):** $\text{Volume} = \text{Cross-sectional Area} \times \text{Length}$

PART 3: VOLUME CALCULATIONS

01. Volume of a Cube

$$\text{Volume} = a \times a \times a = a^3$$

02. Volume of a Cuboid

$$\text{Volume} = a \times b \times c = abc$$

03. Volume of a Pyramid

$$\text{Volume} = \frac{1}{3} \times a \times b \times h = \frac{1}{3} abh$$

04. Volume of a Cone

$$\text{Volume} = \frac{1}{3} \times \pi \times r^2 \times h = \frac{1}{3} \pi r^2 h$$

05. Volume of a Cylinder

$$\text{Volume} = \pi \times r^2 \times h = \pi r^2 h$$

06. Volume of a Sphere

$$\text{Volume} = \frac{4}{3} \pi r^3$$